6 The pick-up on an electric guitar produces an electrical signal from the vibrations of the guitar string. The pick-up consists of a small coil of insulated wire wound round a small cylindrical magnet as shown in Fig. 6.1.

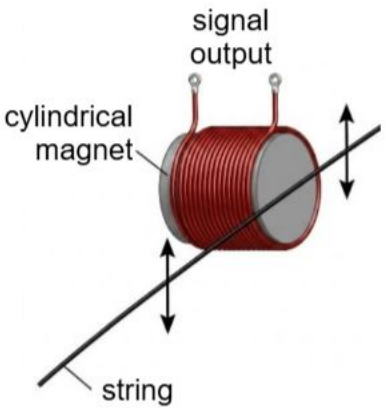


Fig. 6.1

The strings of the guitar are made of steel, which is a conductor. When a string vibrates, an electrical signal is generated between the terminals of the coil.

(a) Use Faraday's Law to explain why an electrical signal is generated.

The string is magnetised by the magnet and becomes a temporary magnet. B1

As the string vibrates, it produces a varying magnetic field and hence the coil experiences a rate of change of magnetic flux linkage. B1

Faraday's law states that induced emf is directly proportional to the rate of change of magnetic flux linkage, B1 thus an emf is induced and a signal in the coil. [3]

(b) Suggest why

(i) the electrical signal needs to be amplified before connection to a loudspeaker,

The amplitude of the vibration is small / the strength of the magnetic field due to the string is weak, hence the induced e.m.f. is small. B1 [1]

(ii) the design of the pick-up would be inappropriate for use on a guitar with nylon strings.

Nylon is non-magnetic material and thus cannot be magnetised. B1

Hence it is unable to produce any rate of change of magnetic flux linkage in the coil and thus no signal generated. B1

[2]

[Total: 6]